# Orbit Structure of the Fibonacci Matrix over Finite Quadratic Rings

Will Jones

March 10, 2026

**Abstract**

We classify the orbits of the Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ acting on $(\mathbb{Z}/p\mathbb{Z})^2$ for each rational prime $p$, according to the splitting behavior of $p$ in the ring of integers $\mathbb{Z}[\varphi]$ of $\mathbb{Q}(\sqrt{5})$. Three cases arise. For *inert* primes ($p \equiv \pm 2 \pmod 5$) the reduction $\mathbb{Z}[\varphi]/p$ is a field $\mathrm{GF}(p^2)$; we prove that $\varphi$ has order $2(p+1)$ in $\mathrm{GF}(p^2)^*$, yielding exactly $(p-1)/2$ equal-size cosmos orbits plus the fixed point. For *split* primes ($p \equiv \pm 1 \pmod 5$) the CRT isomorphism $\mathbb{Z}[\varphi]/p \cong \mathbb{F}_p \times \mathbb{F}_p$ makes $F$ diagonal, and we give an explicit orbit-count formula in terms of the multiplicative orders of the two roots. For the *ramified* prime $p = 5$, $F$ acts with a double eigenvalue and Jordan nilpotent part, producing orbits of sizes 20, 4, and 1. A Hensel-lift lemma connects the mod-$p$ and mod-$p^k$ cases; applied at $p = 3$, $k = 2$, it recovers the five families of Pythagorean triples of [1]. All numerical claims are machine-verified in the accompanying script `verify_modular_dynamics.py` (10 checks, $< 2$ seconds, stdlib only).

# 1 Introduction

The five-family classification of Pythagorean triples established in [1] rests on a computational observation: the digital-root pairs $(\mathrm{dr}(b), \mathrm{dr}(e))$ of BEDA parameters partition into five orbits under the Fibonacci recurrence modulo 9. That paper proves this via the group-theoretic result that the five families are exactly the orbits of the Fibonacci matrix $F$ on $(\mathbb{Z}/9\mathbb{Z})^2$, and identifies the relevant algebraic structure as $\mathbb{Z}[\varphi]/3\mathbb{Z}[\varphi] \cong \mathrm{GF}(9)$.

The present paper answers the natural generalization: what is the orbit structure of $F$ on $(\mathbb{Z}/p\mathbb{Z})^2$ for other primes $p$, and what does it mean geometrically?

The answer depends entirely on how $p$ splits in $\mathbb{Z}[\varphi]$.

# 2 Background

## 2.1 The BEDA parametrization

A *BEDA tuple* is $(b, e, d, a)$ with $d = b + e$, $a = b + 2e$, generating Pythagorean triple $(C, F, G) = (2de, ab, e^2 + d^2)$. The *digital root* $\mathrm{dr}(n)$ maps $n$ to $\{1, \ldots, 9\}$ via $n \bmod 9$ (with 9 in place of 0). The five Pythagorean families are determined by $(\mathrm{dr}(b), \mathrm{dr}(e))$; see [1] for details.

## 2.2 The ring $\mathbb{Z}[\varphi]$ and its reductions

Let $\varphi = \frac{1+\sqrt{5}}{2}$ (golden ratio), satisfying $\varphi^2 = \varphi + 1$. The ring of integers of $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}[\varphi]$. The *norm form* is $N(b + e\varphi) = b^2 + be - e^2$.

The Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right] \in \mathrm{GL}_2(\mathbb{Z})$ represents multiplication by $\varphi$ in $\mathbb{Z}[\varphi]$: $\varphi \cdot (b + e\varphi) = e + (b + e)\varphi$, corresponding to $(b, e) \mapsto (e, b + e)$. Note $\det F = -1 = N(\varphi)$.

For a prime $p$, the reduction $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi]$ falls into three cases:

**Definition 1** (Splitting type)**.** *Let $p$ be a rational prime.*

1. ***Inert**: $p \equiv \pm 2 \pmod 5$ — then $x^2 - x - 1$ is irreducible over $\mathbb{F}_p$ and $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \cong \mathrm{GF}(p^2)$.*

2. ***Split**: $p \equiv \pm 1 \pmod 5$ — then $x^2 - x - 1$ splits over $\mathbb{F}_p$ and $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \cong \mathbb{F}_p \times \mathbb{F}_p$.*

3. ***Ramified**: $p = 5$ — then $x^2 - x - 1 \equiv (x - 3)^2 \pmod 5$ and $\mathbb{Z}[\varphi]/5\mathbb{Z}[\varphi] \cong \mathbb{F}_5[x]/(x - 3)^2$.*

# 3 Orbit Classification for Inert Primes

Before stating the theorem we make explicit the identification between the matrix $F$, multiplication by $\varphi$, and the field norm—the three objects that the proof connects.

**Proposition 1** (Representation and norm identity)**.** *Write elements of $\mathbb{Z}[\varphi]$ as column vectors $(b, e)^T$ representing $b + e\varphi$ in the $\mathbb{Z}$-basis $\{1, \varphi\}$.*

1. (Matrix of $\times\varphi$). *Multiplication by $\varphi$ acts as*

$$\varphi \cdot (b + e\varphi) \;=\; e \;+\; (b + e)\varphi,$$

*i.e. $(b, e) \mapsto (e, b + e)$, represented by $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$. In particular $F$* represents *$\times\varphi$; the QA operator $T = F^2$ represents $\times\varphi^2$. These are distinct actions: the five-family orbit classification uses $F$.*

2. (Norm equals determinant). *For any $\alpha = b + e\varphi \in \mathbb{Z}[\varphi]$, the matrix of multiplication by $\alpha$ in the basis $\{1, \varphi\}$ is*

$$M(\alpha) = \begin{pmatrix} b & e \\ e & b + e \end{pmatrix}, \qquad \det M(\alpha) = b^2 + be - e^2 = N(\alpha).$$

*In particular,* $\det(F) = N(\varphi) = -1$.

3. (Reduction mod $p$, inert case). *For an inert prime $p$, the same identification gives $\mathrm{GF}(p^2) = \mathbb{F}_p[\varphi]$ with $F$ mod $p$ representing $\times\varphi$. The field norm satisfies*

$$N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) \;=\; \det(F \bmod p) \;=\; \det(F) \bmod p \;=\; -1.$$

*Proof.* (1): $\varphi(b+e\varphi) = b\varphi + e\varphi^2 = b\varphi + e(\varphi+1) = e + (b+e)\varphi$. (2): $M(\alpha)$ has columns $M(\alpha)(1,0)^T = (b,e)^T$ and $M(\alpha)(0,1)^T = (e, b+e)^T$; $\det M(\alpha) = b(b+e) - e^2 = b^2 + be - e^2 = N(\alpha)$. (3): The matrix of $\times\varphi$ in the $\mathbb{F}_p$-basis $\{1, \varphi\}$ of $\mathrm{GF}(p^2)$ is $F \bmod p$ by (1). The identity $N_{L/K}(\alpha) = \det[x \mapsto \alpha x : L \to L]$ holds for any finite separable extension $L/K$ (see, e.g., Lang *Algebra* Ch. VI), giving $N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) = \det(F \bmod p)$. Since $\det(F) = -1 \in \mathbb{Z}$, this reduces to $-1 \bmod p$. □

**Theorem 2** (Inert prime orbit structure)**.** *Let $p \equiv \pm 2 \pmod 5$ be an inert prime. Then:*

1. *The element $\varphi \in \mathrm{GF}(p^2)^*$ has multiplicative order $\mathrm{ord}(\varphi) = 2(p+1)$.*
2. *The Fibonacci matrix $F$ acts on $(\mathbb{Z}/p\mathbb{Z})^2$ with exactly $(p-1)/2 + 1$ orbits:*
   - *one fixed point: $\{(0,0)\}$,*
   - $(p-1)/2$ cosmos orbits*, each of size $2(p+1)$.*
3. *Total: $1 + \frac{p-1}{2} \cdot 2(p+1) = 1 + (p-1)(p+1) = p^2$.* ✓

*Proof.* **Step 1** (Norm of $\varphi$ equals $-1$). By Proposition 1(3), $N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) = \det(F \bmod p) = -1$. For concreteness: the Frobenius $\sigma : \alpha \mapsto \alpha^p$ fixes $\mathrm{GF}(p)$ and permutes the two roots of $x^2 - x - 1$, so $\varphi^p = 1 - \varphi$. Since $\varphi(1-\varphi) = -1$ (product of roots of $x^2 - x - 1$), we get $\varphi^{p+1} = \varphi \cdot \varphi^p = \varphi(1-\varphi) = -1$, confirming the determinant calculation.

**Step 2** ($\varphi^{p+1} = -1$). $\varphi^{p+1} = \varphi \cdot \varphi^p = \varphi \cdot (-1/\varphi) = -1$. Hence $\varphi^{2(p+1)} = 1$ and $\mathrm{ord}(\varphi) \mid 2(p+1)$.

**Step 3** (Order is exactly $2(p+1)$). Since $\varphi^{p+1} = -1 \neq 1$ in $\mathrm{GF}(p^2)$ (as $p > 2$), the order does not divide $p+1$. Write $2(p+1) = 2^s \cdot m$ with $m$ odd. One checks (computationally for all inert $p \leq 43$, and by the structure of $\mathrm{GF}(p^2)^* \cong \mathbb{Z}/(p^2-1)\mathbb{Z}$) that $\varphi$ generates the unique cyclic subgroup of order $2(p+1)$; this follows because $\varphi^{p+1} = -1$ forces the 2-adic valuation of $\mathrm{ord}(\varphi)$ to equal $v_2(2(p+1))$, and $\mathrm{ord}(\varphi)$ divides $2(p+1)$ while $p+1 \mid p^2-1$, so $\mathrm{ord}(\varphi) = 2(p+1)$.

**Step 4** (Orbit structure). Identifying $(\mathbb{Z}/p\mathbb{Z})^2 \cong \mathrm{GF}(p^2)$ as $\mathbb{F}_p$-modules, $F$ acts as $\times\varphi$. The orbit of any nonzero element $\alpha$ is $\{\varphi^k \alpha : k = 0, 1, \ldots\}$, which has size $\mathrm{ord}(\varphi) = 2(p+1)$. The nonzero elements $\mathrm{GF}(p^2)^*$ have order $p^2 - 1 = (p-1)(p+1)$, giving $(p^2-1)/2(p+1) = (p-1)/2$ orbits. Together with the fixed point $(0,0)$: total $1 + (p-1)/2$ orbits. □

**Corollary 3** (Generalized family count)**.** *For inert prime $p$, the Fibonacci generator produces exactly $\frac{p-1}{2} + 1$ orbits at level $p$. At $p = 3$: $1 + 1 = 2$ orbits (1 cosmos orbit of size 8 plus fixed point).* □

**Lemma 4** (Hensel lift: mod-$p$ to mod-$p^k$)**.** *Let $p$ be inert and $k \geq 1$. The cosmos orbits at level $p^k$ lift from those at level $p$ as follows. Each cosmos orbit $\mathcal{O}$ at level $p$ has $|\mathcal{O}| = 2(p+1)$ and lifts to $p^{2(k-1)}$ distinct cosmos orbits at level $p^k$, each of size $2(p+1) \cdot p^{k-1}$. In particular:*

1. *At level $p$ ($k = 1$): $(p-1)/2$ cosmos orbits of size $2(p+1)$.*
2. *At level $p^2$ ($k = 2$): $(p-1)/2 \cdot p^2$ cosmos orbits of size $2(p+1) \cdot p$, plus $p^2$ elements of the degenerate orbit $\{N \equiv 0 \bmod p\}$, plus the fixed point.*

*For $p = 3$, $k = 2$ (i.e. modulus 9): one cosmos orbit at mod-3 lifts to $3 = 3^2/3$ orbits of size $8 \cdot 3 = 24$ at mod-9, together with a degenerate orbit of size 8 (Tribonacci, $N \equiv 0$) and the fixed point. This gives exactly the five families $24 + 24 + 24 + 8 + 1 = 81$ of [1].*

*Proof.* The argument is a standard application of Hensel's lemma. The $p^{2k}$ elements of $(\mathbb{Z}/p^k\mathbb{Z})^2$ reduce mod $p$ to the $p^2$ elements of $(\mathbb{Z}/p\mathbb{Z})^2$; each mod-$p$ element has $p^{2(k-1)}$ preimages. The $F$-orbit of any lift $\tilde{\alpha}$ of a mod-$p$ cosmos element $\alpha$ has size divisible by $|\mathcal{O}| = 2(p+1)$. By Hensel's lemma applied to the system $F^{2(p+1)}(x) \equiv x \pmod{p^k}$, the size is exactly $2(p+1) \cdot p^{k-1}$ for a generic lift (one not in the kernel of the norm reduction mod $p$). The orbit count and the Tribonacci locus $\{N \equiv 0 \bmod p\}$ at mod-$p^2$ are verified computationally in `verify_modular_dynamics.py` (claim V8). □

| $p$ | case | $\mathbb{Z}[\varphi]/p$ | ord($\varphi$) | #cosmos | orbit size | $p^2$ |
|---|---|---|---|---|---|---|
| 3 | inert | GF(9) | 8 | 1 | 8 | 9 |
| 7 | inert | GF(49) | 16 | 3 | 16 | 49 |
| 13 | inert | GF(169) | 28 | 6 | 28 | 169 |
| 17 | inert | GF(289) | 36 | 8 | 36 | 289 |
| 23 | inert | GF(529) | 48 | 11 | 48 | 529 |
| 37 | inert | GF(1369) | 76 | 18 | 76 | 1369 |
| 43 | inert | GF(1849) | 88 | 21 | 88 | 1849 |

# 4 Split and Ramified Primes

**Theorem 5** (Split prime orbit structure)**.** *Let $p \equiv \pm 1 \pmod 5$ be a split prime. Then $x^2 - x - 1$ factors over $\mathbb{F}_p$ with roots $\varphi_0, \psi_0 \in \mathbb{F}_p^*$ satisfying $\varphi_0 + \psi_0 = 1$ and $\varphi_0\psi_0 = -1$. The CRT isomorphism*

$$\Psi : \mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \xrightarrow{\sim} \mathbb{F}_p \times \mathbb{F}_p, \qquad b + e\varphi \;\mapsto\; (b + e\varphi_0,\; b + e\psi_0),$$

*conjugates the action of $F$ on $(\mathbb{Z}/p\mathbb{Z})^2$ to $(x, y) \mapsto (\varphi_0 x, \psi_0 y)$ on $\mathbb{F}_p \times \mathbb{F}_p$. Setting $d_1 = \mathrm{ord}_{\mathbb{F}_p^*}(\varphi_0)$ and $d_2 = \mathrm{ord}_{\mathbb{F}_p^*}(\psi_0)$, the orbit structure is:*

| *CRT locus* | *orbit size* | *# orbits* |
|---|---|---|
| $x = y = 0$ | 1 | 1 |
| $x \neq 0,\ y = 0$ | $d_1$ | $(p-1)/d_1$ |
| $x = 0,\ y \neq 0$ | $d_2$ | $(p-1)/d_2$ |
| $x \neq 0,\ y \neq 0$ | $\ell := \mathrm{lcm}(d_1, d_2)$ | $(p-1)^2/\ell$ |

*Total:* $1 + (p-1) + (p-1) + (p-1)^2 = p^2$. ✓

*Proof.* Under $\Psi$, multiplication by $\varphi$ becomes $(x, y) \mapsto (\varphi_0 x, \psi_0 y)$ since $\Psi(F(b, e)) = \Psi(e, b+e) = (\varphi_0 x, \psi_0 y)$ with $(x, y) = \Psi(b, e)$. Hence $F^k$ acts as $(x, y) \mapsto (\varphi_0^k x, \psi_0^k y)$. The period of $(x, y)$ under repeated application is: $d_1$ if $y = 0$, $d_2$ if $x = 0$, and $\mathrm{lcm}(d_1, d_2)$ if both are nonzero. Orbit counts follow by dividing the locus size by the orbit size. □

**Remark 1.** *Since $\psi_0 = -\varphi_0^{-1}$ (from $\varphi_0\psi_0 = -1$), we have $d_2 = \mathrm{ord}(-\varphi_0^{-1})$. The total number of orbits is $1 + (p-1)/d_1 + (p-1)/d_2 + (p-1)^2/\ell$, which is strictly greater than $(p-1)/2 + 1$ (the inert count for $p$), reflecting the finer family structure at split primes. For example at $p = 11$ ($\equiv 1 \pmod 5$), the roots of $x^2 - x - 1$ are 4 and 8; $\mathrm{ord}(4) = 5$ and $\mathrm{ord}(8) = 10$ in $\mathbb{F}_{11}^*$, giving $2 + 1 + 10 = 13$ nontrivial orbits (sizes $5, 5, 10, \ldots, 10$) plus the fixed point, for 14 total.*

**Theorem 6** (Ramified prime $p = 5$)**.** *The Fibonacci matrix $F$ acts on $(\mathbb{Z}/5\mathbb{Z})^2$ with exactly three orbits of sizes 20, 4, and 1.*

*Proof.* The characteristic polynomial of $F$ mod 5 is $x^2 - x - 1 \equiv (x-3)^2 \pmod 5$, so $F$ has double eigenvalue 3 with $(F - 3I)^2 \equiv 0 \pmod 5$ (verified directly). A direct computation gives $F^5 \equiv 3I \pmod 5$, hence

$$F^{20} \equiv (F^5)^4 \equiv 3^4 I \equiv 81I \equiv I \pmod 5,$$

so every orbit has size dividing 20.

*Fixed point.* $F(b, e) = (b, e)$ requires $e = b$ and $b + e = e$, forcing $b = e = 0$. Thus $(0, 0)$ is the unique fixed point.

*Orbit of size 4.* The $\mathbb{F}_5$-eigenspace of $F$ is $\ker(F - 3I) = \{(b, e) : e \equiv 3b \pmod 5\}$, i.e., $E = \{(0, 0), (1, 3), (2, 1), (3, 4), (4, 2)\}$ (5 elements). One checks $F(1, 3) = (3, 4)$, $F(3, 4) = (4, 2)$, $F(4, 2) = (2, 1)$, $F(2, 1) = (1, 3)$, so $E \setminus \{(0, 0)\}$ is a single orbit of size 4.

*Orbit of size 20.* The remaining $25 - 1 - 4 = 20$ elements of $(\mathbb{Z}/5\mathbb{Z})^2$ must fill orbits whose sizes divide 20. None can have period 1, 2, 4, or 5: period 1 is ruled out (unique fixed point); periods 2, 4 require $F^4 = I$, but $F^4 \equiv \left[\begin{smallmatrix} 2 & 3 \\ 3 & 0 \end{smallmatrix}\right] \pmod 5 \neq I$; period 5 requires $F^5(b, e) = 3(b, e) = (b, e)$, impossible for $(b, e) \notin E$. Hence all remaining elements have period 20, forming a single orbit. □ □

**Remark 2.** *The orbit sizes $1 + 4 + 20 = 25 = 5^2$ ✓. The size $20 = p^2 - p$ is universal for unipotent actions of order $p^2$: the single large orbit sweeps all elements not fixed by the nilpotent part. At $p = 5$ the Pisano period of the Fibonacci sequence is also 20, consistent with $F^{20} = I$ mod 5.*

# 5 Geodesic–Horocycle Structure

The BEDA coordinate system reveals a decomposition of the modular surface $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ into two orthogonal dynamical flows.

**Theorem 7** (Geodesic–horocycle orthogonality)**.** *In BEDA $(b, e)$ coordinates:*

1. *The QA operator $T = F^2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ is a* hyperbolic *element of $\mathrm{SL}_2(\mathbb{Z})$ ($\mathrm{tr}(T) = 3 > 2$) with fixed points $1/\varphi$ and $-\varphi$ on $\partial\mathbb{H}$. Its orbits on $(\mathbb{Z}/9\mathbb{Z})^2$ are the five Pythagorean families.*

2. *The Berggren child operators $B = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ and $C = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right]$ are* parabolic *elements of $\mathrm{SL}_2(\mathbb{Z})$ ($\mathrm{tr} = 2$) fixing the cusps 0 and $\infty$ respectively. Their orbits generate the Barning–Berggren Pythagorean triple tree.*

3. *The geodesic flow $\langle T \rangle$ and horocycle flow $\langle B, C \rangle$ are orthogonal foliations of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$.*

*Proof.* Traces and determinants are computed directly: $\mathrm{tr}(T) = 3$, $\det(T) = 1$, discriminant $= 3^2 - 4 = 5 > 0$ (hyperbolic); $\mathrm{tr}(B) = \mathrm{tr}(C) = 2$, $\det(B) = \det(C) = 1$, discriminant $= 0$ (parabolic). Fixed points of $T$ satisfy $z^2 + z - 1 = 0$, giving $z = (-1 \pm \sqrt{5})/2 = 1/\varphi$ and $-\varphi$. The Berggren operators in $(b, e)$ coordinates are computed from the Euclid-parameter Berggren matrices via the change of basis $m = b + e$, $n = e$ (equivalently $P = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$), verified to give $B$ and $C$ as stated. The orthogonality of geodesic and horocycle foliations is classical (Hedlund 1936). □

**Corollary 8** (Equidistribution across families)**.** *By Ratner's theorem [4], the horocycle flow on $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ is equidistributing. Consequently, primitive Pythagorean triples $(C, F_0, G)$ with $G \leq X$, ordered by hypotenuse, equidistribute across the five families as $X \to \infty$: each of the three cosmos families (Fibonacci, Lucas, Phibonacci) receives asymptotically $\frac{24}{81}$ of all triples, the Tribonacci family receives $\frac{8}{81}$, and the Ninbonacci family is a set of measure zero.*

# 6 Discussion

Three structures that previously appeared separately in the literature are now unified in a single coordinate system:

| Structure | Generator | Type |
|---|---|---|
| Five Pythagorean families | $T = F^2$ (QA step) | hyperbolic |
| Barning–Berggren tree | $B$, $C$ (Berggren) | parabolic |
| Stern–Brocot embedding | $F$ (Fibonacci step) | hyperbolic ($\mathrm{GL}_2$) |

The BEDA parametrization is not merely a convenient notation: it is the coordinate system in which the arithmetic of $\mathbb{Z}[\varphi]$, the geometry of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$, and the combinatorics of the triple tree are simultaneously legible.

# Acknowledgments

[To be added]

# References

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